

CS 540 Midterm Sample Questions

1. Which of the following sentence is FALSE regarding regression?
 - (a) It discovers causal relationships
 - (b) It relates inputs to outputs.
 - (c) It is used for prediction.
 - (d) It may be used for interpretation.

2. Consider the following statements regarding k -nearest neighbors.
 - I. If point A is among point B's k -nearest-neighbors, then B is among A's k -nearest-neighbors.
 - II. On a training set consisting of 100 positive items and 100 negative items, it is possible for 1NN to have training set accuracy 0 (i.e., it can completely fail on the training set).
 - (a) I is True, II is False
 - (b) I is False, II is True
 - (c) Both are True
 - (d) Both are False

3. What assumption does naive Bayes make?
 - (a) Conditional independence of features
 - (b) Conditional dependence of features
 - (c) Independence of features
 - (d) Dependence of features

4. What is the primary difference between supervised and unsupervised learning? Please choose all that apply
 - (a) Supervised learning gives you what some expected outputs are, unsupervised learning does not
 - (b) There is very little difference, they are the same in most contexts
 - (c) Supervised learning is often used for prediction tasks, but unsupervised learning is usually not
 - (d) Unsupervised learning produces correct labels rather than being given correct labels

5. Consider the following dataset about 5 types of mushrooms, about whether the mushroom

is poisonous based on three binary features. We want to use the naive Bayes method to classify a new mushroom with Height=Tall, Body=Slim, and Color=Black. What's the probability of this mushroom being poisonous?

Height	Body	Color	Poisonous
Tall	Fat	Black	No
Tall	Fat	White	Yes
Short	Slim	White	No
Short	Slim	Black	Yes
Tall	Slim	White	Yes

- (a) 0.64
- (b) 0.82
- (c) 0.38
- (d) 0.67

6. We are given data consisting of the following one-dimensional points 1, 3, 5, 6, 7, 12, 13, 15. What will be the first pair of clusters in 2-mean clustering given the initial centers 1, 12?

- (a) {1, 3, 5, 6, 7}, {12, 13, 15}
- (b) {1, 3, 5, 6}, {7, 12, 13, 15}
- (c) {1, 3}, {5, 6, 7, 12, 13, 15}
- (d) {1, 3, 5, 6, 7, 12}, {13, 15}

7. We have the following one dimensional dataset that has been merged into three clusters: $C1 = \{1, 2\}$, $C2 = \{4, 5\}$, $C3 = \{6, 10\}$. Which clusters will be merged next using single linkage? Which will be merged using complete linkage?

- (a) C2 and C3, C1 and C2
- (b) C1 and C2, C2 and C3
- (c) C1 and C3, C2 and C3
- (d) C2 and C3, C1 and C3

8. Suppose we have the following training corpus in an NLP setup. "I took my dog out for a walk but my dog ran away" (i) What is the $P(\text{dog})$ using a unigram model without Laplace smoothing? (b) What is the $P(\text{my dog})$ using a unigram model without Laplace smoothing? (c) What is the $P(\text{my dog})$ using a bigram model without Laplace smoothing?

Please show your work for each part.

9. Suppose you are performing PCA in $\mathbb{R}^3$ (3-dimensional space). It turns out that the first principal component is $v_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ and the second principal component is $v_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$. One of the original data points is $x = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$. What will its new representation be after projection?

10. You have a dataset for regression given by $(x_1, y_1) = \left(\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, 2 \right)$, $(x_2, y_2) = \left(\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, 4 \right)$.

(a) What are the labels? What is the number of points n and the dimension of the feature vectors d ?

(b) We have the weights $\beta_0 = 0, \beta_1 = 2, \beta_2 = 1, \beta_3 = 1$. Predict $\hat{y}$ for $x = \begin{bmatrix} 1 \\ 10 \\ 1 \end{bmatrix}$.

(c) For the weights above, compute the mean squared error.

11. Suppose we want to solve the following k -class classification problem with cross entropy loss $\ell(\mathbf{y}, \hat{\mathbf{y}}) = -\sum_{j=1}^k y_j \log \hat{y}_j$, where the ground truth and predicted probabilities $\mathbf{y}, \hat{\mathbf{y}} \in \mathbb{R}^k$. Recall that the softmax function turns the output into probabilities: $\hat{y}_j = \frac{\exp f_j(x)}{\sum_i^k \exp f_i(x)}$. A diagram is provided below. What is the partial derivative $\partial_{f_j} \ell(\mathbf{y}, \hat{\mathbf{y}})$?

For notational simplicity, we use y_i to denote $\mathbf{1}\{y_i = 1\}$, and $\hat{y}_i$ as $p(y_i = 1 | \mathbf{x}; \theta)$.

